ERES INSTITUTE FOR NEW AGE CYBERNETICS

SECURITY, PRIVACY AND TRUST

[DRAFT]

MIEVM-Consolidated Specification

Anchored in M×E+C=R — Matter × Energy + Constant = Reason
with United States Constitutional Framing, Economic Integration,
and Comprehensive A–Z Numerics Index

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Version 0.9 DRAFT — MIEVM Consolidated — March 2026
Integrating: IDIPITIS · FAVORS · CBGMODD · Gracechain · SECUIR · NBERS
MIEVM Nodes: Claude (Anthropic) · Grok (xAI) · DeepSeek

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“Don’t hurt yourself. Don’t hurt others. Build for generations to come.”

DRAFT — MIEVM Consolidated. This document synthesizes parallel validation from Claude (Anthropic), Grok (xAI), and DeepSeek. Architectural intent declared; implementation annexes pending. Final normative status contingent upon prototype validation per the Patience Protocol.

MIEVM PROVENANCE RECORD

This consolidated specification was produced through the Multi-Instrument Ensemble Validation Method (MIEVM), employing three independent AI systems as parallel analytical nodes. The following record documents the validation chain and the specific contributions of each instrument:

Node	Instrument	Contribution
Node 1	Claude (Anthropic)	Primary drafting: 12-section architectural specification, SPT Triad Integrity Condition, unified threat model, subsystem specifications (IDIPITIS, FAVORS, CBGMODD, Gracechain, SECUIR, NBERS), degraded state management, dependency graph. Live-session correction: Erasure Completeness Principle (§6.4), co-developed with JAS, replacing binary proof-of-erasure with thermodynamically honest ECR metric.
Node 2	Grok (xAI)	Forensic engineering critique (6.1/10): Identified gaps in formal rigor, absence of prior art engagement, implementation complexity concerns, circular dependency risk analysis. Provided credits, references, and CCAL license expansion.
Node 3	DeepSeek	Philosophical-architectural analysis (9.0/10): Validated moral vector, triune relationship, threat model innovation (Classes III and V), subsystem integration coherence. Produced meta-review comparing Node 2 analysis. Authored Patience Protocol establishing versioning discipline.

MIEVM Convergence Assessment: All three nodes converge on the document's philosophical strength, the originality of the triune integrity condition, and the innovation of Threat Classes III (Trust Erosion) and V (Epistemic Manipulation). Divergence exists on implementability (Node 2: 3/10 vs. Node 3: acknowledged as next-phase challenge). This divergence is productive—it maps the work that lies between architectural intent and engineering specification. A post-consolidation live-session correction by the primary author (JAS) identified a foundational flaw that all three nodes missed: the

binary framing of proof-of-erasure. The Erasure Completeness Principle (§6.4) replaces this with a thermodynamically honest continuous metric, representing the first original contribution generated through human-MIEVM dialectic rather than parallel validation.

ANCHORING DECLARATION: M×E+C=R

This specification is anchored in the second canonical formula of the ERES Triune Math:

$$M \times E + C = R$$

Matter × Energy + Constant = Reason

The anchoring is not decorative. The formula serves as the organizing principle of the entire SPT architecture. The material substrate of security (M—infrastructure, hardware, biometrics), energized by the operational processes of privacy protection (E—consent management, provenance tracking, encryption), plus the immutable constants of the ERES moral vector (C—the three governing principles, non-negotiable), produce Reason (R)—the measurable, auditable, defensible basis for trust.

Every subsystem specification in this document maps to one or more terms of this formula. SECUIR and VERTECA are Matter-domain systems. FAVORS and CBGMODD are Energy-domain systems. The Three Governing Principles and Triune Math are the Constants. And the BERA indices—ARI, ERI, RHC, RCI—are the instruments that measure whether Reason has, in fact, been produced.

This anchoring also carries an economic implication: Matter and Energy are not free. They require resources. The SPT architecture therefore cannot be designed in isolation from the economic architecture of NAC—specifically UBIMIA, Meritcoin, Gracechain, and SROC—which provides the resource allocation mechanisms through which Matter is procured and Energy is sustained. This integration is specified in Section 8.

1. FOUNDATIONAL DEFINITIONS

The following definitions govern all provisions within this specification. Where these definitions conflict with colloquial or industry-standard usage, the SPT definitions prevail within the NAC context.

1.1 Security

Security is the condition in which a system's integrity boundaries are maintained against all classes of adversarial action—unauthorized access, data exfiltration, identity impersonation, infrastructure compromise, and epistemic manipulation. Within the NAC framework, security is not a static perimeter but a dynamic, self-healing property maintained through continuous cybernetic feedback. Security is measured via the BERA Adaptive Resilience Index (ARI) and is operationalized through IDIPITIS clearance protocols and SECUIR infrastructure provisions.

1.2 Privacy

Privacy is the sovereign right of every individual to determine the scope, duration, audience, and purpose of disclosure for all personally identifying information, biometric data, behavioral telemetry, and deliberative reasoning. Privacy is not the absence of data collection but the presence of enforceable consent architecture. Within NAC, privacy is operationalized through FAVORS biometric sovereignty provisions and CBGMODD data-chain provenance tracking, and is measured via the BERA Resonant Harmony Cycle (RHC) as a community-level indicator of informational self-determination.

1.3 Trust

Trust is a systemic property—not a subjective sentiment—that emerges when security and privacy are simultaneously maintained over time, and when institutions demonstrate verifiable adherence to stated commitments. Trust within NAC is continuously earned, measured, and validated. It is operationalized through Gracechain credentialing, quantified via the BERA Resonant Continuity Index (RCI = P_Ω_norm × ARI_sys × VibConst), and governed by the transparency provisions of the ERES epistemology.

1.4 The SPT Triad Integrity Condition

Security, Privacy, and Trust are in a state of triad integrity when and only when all three conditions are simultaneously satisfied:

1. **ARI ≥ ARI_floor(tier)** — Security is maintained at or above the minimum Adaptive Resilience Index for the applicable GSSG tier.
2. **RHC ∈ [μ – 1σ, μ + 1σ]** — Privacy conditions maintain community Resonant Harmony within one standard deviation of the rolling mean.
3. **RCI ≥ 0.7 × RCI_baseline** — Systemic trust has not degraded below 70% of the established baseline for the domain in question.

When any condition fails, the SPT architecture enters a degraded state per Section 10.

2. UNIFIED THREAT MODEL

The SPT operates against a unified threat model recognizing five classes of adversarial action. These classes are not mutually exclusive; sophisticated attacks typically span multiple classes simultaneously.

Class	Designation	Description and Countermeasure Domain
I	**IDENTITY COMPROMISE**	Unauthorized creation, duplication, impersonation, or degradation of identity credentials. Includes synthetic identity generation, biometric spoofing, credential theft,

		clearance escalation. Countermeasure: IDIPITIS + FAVORS.
II	**DATA EXFILTRATION**	Unauthorized extraction, duplication, or transmission of protected data. Includes bulk harvesting, side-channel leakage, coerced disclosure, supply-chain interception. Countermeasure: CBGMODD + SECUIR.
III	**TRUST EROSION**	Deliberate degradation of institutional trustworthiness without breaching security/privacy perimeters. Includes disinformation, algorithmic bias injection, selective transparency manipulation, credential inflation. Countermeasure: Gracechain + MIEVM.
IV	**INFRASTRUCTURE DISRUPTION**	Physical or digital disruption of the material substrate. Includes energy grid attacks, facility compromise, backbone severance, DDoS against SOMT/GAIA channels. Countermeasure: SECUIR + VERTECA.
V	**EPISTEMIC MANIPULATION**	Attacks against knowledge-production and decision-making processes. Includes MIEVM instrument corruption, BERA input poisoning, SCALULAR assessment manipulation, false premise insertion into Public Reasoning Archives. Countermeasure: MIEVM + CBGMODD + PlayNAC.

3. IDIPITIS: IDENTITY INTEGRITY AND SECURITY CLEARANCE

IDIPITIS is the identity and security clearance subsystem—the primary boundary-enforcement mechanism determining who accesses what, under which conditions, with what degree of accountability.

3.1 Identity Lifecycle

- **Genesis.** Initial identity creation through NBERS enrollment. Foundational biometric data processed through FAVORS; SSSC Tier 1 credential issued via SCALULAR; unique IDIPITIS identity vector assigned. No identity without a verifiable biological person.
- **Maturation.** Progressive credential accumulation through EarnedPath milestones, SCALULAR certifications (SSHP, SSLA, SSPS, SSST), and Gracechain merit accrual. Each credential layer expands access and increases accountability surface.
- **Active Operation.** Steady-state: identity continuously validated against FAVORS biometric checkpoints, BERA indices maintained within acceptable ranges, full credential-authorized access rights exercised.

- **Suspension.** Temporary privilege revocation triggered by BERA degradation, security investigation, or voluntary self-suspension. Non-punitive by default.
- **Archival.** Terminal state preserving identity in permanent record for historical continuity, estate adjudication, and generational knowledge transfer. Read-only access to own Public Reasoning Archive contributions.

3.2 Security Clearance Architecture

IDIPITIS implements a graduated clearance architecture mapped to the GSSG tier system. Clearance is a continuous variable—a function of credential stack, current BERA indices, biometric verification freshness, and resource sensitivity classification:

```
CLR_eff = f(CRED_stack, ARI_current, BIO_fresh, RES_class)
```

Access is granted when CLR_eff exceeds the resource's minimum clearance threshold. Denial includes transparent explanation of which component(s) fell short. This represents a fundamental departure from static role-based access control toward dynamic, context-aware identity management.

4. TRIUNE MATH CONSTRAINT ARCHITECTURE

All SPT operations, resolutions, and remediation actions are validated against the three canonical formulas of the ERES Triune Math, which serve as boundary conditions ensuring that no action within the system exceeds the resources available, produces unreasonable outcomes, or is unimplementable in its spatiotemporal context.

4.1 The Three Constraints

- **C = R×P/M** — Cybernetics = Resource × Purpose ÷ Method. Actions must allocate resources proportionate to stated purpose; implementation method must not exceed the resource-purpose ratio by more than 15%.
- **M×E+C = R** — Matter × Energy + Constant = Reason. Material and energetic costs plus systemic constants must produce reasonable outcomes. Failures classified as "unsound." This is the anchoring formula of the present specification.
- **REAL = (E·M·R)/(T·S)** — Energy · Matter · Resonance over Time · Space. Real-world impact assessed against spatiotemporal implementation constraints. Values below the tier-appropriate floor indicate unimplementability.

4.2 BERA Index Thresholds

Index	Full Name	SPT Minimum Threshold
ARI	Adaptive Resilience Index	No party's ARI reduced below pre-dispute baseline by >5%. Net system ARI non-negative.

ERI	Economic Resilience Index	Positive or neutral ERI trajectory within one UBIMIA distribution cycle. No new dependency cascades.
RHC	Resonant Harmony Cycle	Community RHC within one standard deviation of pre-action mean.
RCI	Resonant Continuity Index	RCI = P_Ω_norm × ARI_sys × VibConst. Must not fall below 0.7 of pre-action value. Below 0.7 triggers automatic GSSG tier escalation.

5. FAVORS: BIOMETRIC SOVEREIGNTY

FAVORS is the biometric subsystem. Its fundamental principle: an individual's biometric data belongs to that individual in the same ontological category as their body. Biometric data is not "collected"; it is "entrusted" under conditions the individual defines and can revoke.

5.1 Seven-Layer Consent Architecture

Layer	Designation	Governance
1	**COLLECTION**	Explicit consent before any capture. Must specify modality, resolution, purpose. Blanket consent prohibited.
2	**STORAGE**	Individual designates jurisdiction, encryption standard, retention duration. Default: encrypted at rest, home GSSG tier, 10-year retention with mandatory re-consent.
3	**PROCESSING**	Each computational operation requires purpose-specific authorization. Logs immutable via CBGMODD provenance chains.
4	**MATCHING**	Verification events governed separately from raw processing. Binary result only; raw data never transmitted to verifier.
5	**SHARING**	Cross-system/cross-jurisdictional transmission requires explicit, granular, time-bounded consent. Logged on Gracechain.
6	**DERIVATION**	Derivative products (templates, hashes, behavioral models) require separate consent. Derivatives inherit source constraints.
7	**DELETION**	Irrevocable right to demand maximized deletion of all biometric data including derivatives. Verified via CBGMODD erasure completeness audit (ECR). Residual report and containment status logged on Gracechain. See §6.4.

5.2 Anti-Surveillance Provision

No entity operating within the NAC framework—governmental, corporate, or institutional—may conduct passive biometric surveillance. This provision is absolute and non-waivable. Emergency provisions under the Resolution Protocol may compress timelines but may not suspend the anti-surveillance provision. This represents what MIEVM Node 2 (Grok) assessed as “unusually individual-sovereignty-oriented for governance-oriented architectures” and “meaningfully stronger than most government or large corporate privacy frameworks.”

6. CBGMODD: DATA-CHAIN INTEGRITY AND PROVENANCE

CBGMODD provides cryptographically verifiable provenance chains for every datum that enters, traverses, or exits any NAC-governed system. Where FAVORS governs consent over biometric data specifically, CBGMODD governs integrity and traceability of all data universally.

6.1 Provenance Chain Architecture

Every data object carries an immutable provenance chain—a cryptographically signed record of origin, every transformation, every system access, and every consent authorization. Provenance chains are structured as directed acyclic graphs (DAGs) with each node representing a data state and each edge representing a transformation or access event. The chain is signed at each transition using IDIPITIS credentials, creating a multi-party attestation trail.

6.2 Proof-of-Erasure

When an individual exercises FAVORS Layer 7 (Deletion), CBGMODD generates a cryptographic verification of the erasure process across all identified storage locations including backups, caches, and derivatives. The erasure audit is logged on Gracechain and independently auditable. Section 6.4 formally corrects the scope of this claim: erasure is asymptotic, not binary, and CBGMODD’s role is reconceived as an erasure completeness auditor producing an Erasure Completeness Ratio (ECR) rather than a binary certificate.

6.3 CBGMODD and the Public Reasoning Archive

CBGMODD provides the integrity backbone for the Public Reasoning Archive specified in the Resolution Protocol (ERES-RESPROTOCOL-2026-001). Every Archive entry carries a full provenance chain, ensuring the deliberative record cannot be retroactively altered, selectively redacted, or falsely attributed. This is the technical instantiation of the ERES epistemological principle that shared reasoning constitutes shared wealth.

6.4 The Erasure Completeness Principle

6.4.1 The Thermodynamic Boundary

The preceding sections specify a “proof-of-erasure” protocol as though erasure were a binary condition: data is either gone or it is not. This section formally corrects that formulation. Complete erasure of information is, in the general case, impossible—for the same fundamental reason that complete recycling of matter is impossible. Both encounter the thermodynamic boundary: entropy cannot be fully reversed within a finite energy budget. There is always residual signal—magnetic remnance on a storage platter, cached fragments in a CDN node not enumerated in the propagation graph, a derivative hash that outlived its source, a neural trace in a human being who read the data before deletion was requested.

The honest engineering position, therefore, is not “we can prove the data is gone” but rather “we can prove we have maximized the removal, we can measure how close to totality we achieved, and we can specify what residual remains under what containment.” The problem shifts from binary (erased / not erased) to continuous (erasure completeness ratio), and the governance question becomes: what is the acceptable floor for that ratio, and what monitoring regime sustains it over time?

6.4.2 The Erasure Completeness Ratio

The SPT specification introduces the Erasure Completeness Ratio (ECR) as a formally defined metric:

```
ECR = N_confirmed_destroyed / N_total_propagation_graph
```

Where N_confirmed_destroyed is the number of confirmed-eliminated instances of the target data (including copies, caches, backups, and derivative products) and N_total_propagation_graph is the total number of instances identified by CBGMODD’s provenance tracking across the data object’s full lifecycle. ECR is a value between 0 and 1 that approaches but, in the general case, never reaches 1.

The ECR is bounded by the REAL constraint:

```
ECR_max = f(E, M, R, T, S)
```

Erasure effectiveness is bounded by the energy (E) and material resources (M) that can be applied, modulated by the resonance (R) of the system’s coordination capacity, across the time (T) and spatial extent (S) over which the data has propagated. You cannot spend infinite energy chasing the last fragment across infinite space. You maximize within your resource envelope. The Triune Math formula REAL = (E·M·R)/(T·S) provides the formal ceiling: when the marginal cost of pursuing the next data fragment exceeds the REAL value of the system, you have reached the practical limit of erasure.

6.4.3 From Binary Certificate to Completeness Audit

This reframes CBGMODD’s proof-of-erasure role. Instead of issuing a binary certificate (“erased”), CBGMODD becomes an erasure completeness auditor. For each deletion action, CBGMODD:

4. **Maps the propagation graph.** Identifies every known instance of the target data: origin copy, replicas, backups, caches, derivative products (hashes, templates, behavioral models), and any cross-system transmissions logged under FAVORS Layer 5 (Sharing) consents.
5. **Executes confirmed destruction.** Systematically eliminates each identified instance, cryptographically verifying each destruction event.
6. **Calculates ECR.** Divides confirmed destructions by total graph size to produce the Erasure Completeness Ratio.
7. **Reports residual.** Identifies any instances that could not be confirmed destroyed, specifying their location, classification (known-but-inaccessible, suspected-but-unconfirmed, derivative-of-unknown-scope), and the containment measures applied.
8. **Logs to Gracechain.** The Gracechain trust transaction entry does not say "erased." It says: "Erasure completeness: [ECR value], with [percentage] residual in [identified locations], under [specified containment], assessed on [date]." Honest. Measurable. Auditable.

This is the epistemic posture the ERES framework demands: do not claim more than you know. Do not pretend completeness when you have achieved maximization. The distinction matters—both legally and morally.

6.4.4 The Environmental Integrity Analogy

The deeper principle at work here extends beyond data erasure to a civilization-scale observation: the entropy of information mirrors the entropy of matter. Both require continuous expenditure of energy to maintain order. Both degrade without active stewardship. Both demand that governance systems be designed for ongoing maintenance rather than one-time declarations.

A watershed is not "clean" because someone issued a certificate saying it was clean on a particular date. A watershed is clean because a continuous monitoring regime measures contaminant levels, tracks pollution sources, reports honestly on what residual contamination exists, and applies ongoing remediation effort proportionate to the environmental integrity standard being maintained. The same is true of a data ecosystem governed by CBGMODD.

This reframes the entire SPT architecture. The system is not a lock installed once; it is a garden tended indefinitely. Security is not a wall built and walked away from; it is a living boundary continuously maintained. Privacy is not a door closed once; it is a consent architecture continuously honored. Trust is not a credential issued once; it is a relationship continuously earned. The Erasure Completeness Principle makes this continuous-maintenance posture explicit in the data integrity domain, but the principle applies across all six SPT subsystems.

6.4.5 ECR Minimum Thresholds

The following minimum ECR thresholds are established for FAVORS Layer 7 (Deletion) actions:

Data Classification	ECR Floor	Governance
Tier 1 Personal Biometric	**≥ 0.997**	Highest priority. All known propagation nodes must be addressed. Residual limited to statistically unidentifiable fragments. Re-audit at 90 days.
Tier 2 Credentialing Data	**≥ 0.990**	High priority. Derivative credentials may persist in archival form on Gracechain as anonymized trust events. Re-audit at 180 days.
Tier 3 Behavioral Telemetry	**≥ 0.985**	Moderate priority. Aggregate statistical derivatives may persist if fully anonymized beyond re-identification threshold. Re-audit at 365 days.
Tier 4 Institutional Records	**≥ 0.970**	Standard priority. Public Reasoning Archive entries containing the individual's contributions are redacted rather than deleted, preserving deliberative integrity while removing identifying data. Re-audit at 365 days.

ECR values below the applicable floor trigger AMBER degradation (Section 10.1) in the CBGMODD subsystem. Persistent failure to meet ECR floors within the re-audit cycle triggers ORANGE. The individual is notified of all ECR results and residual reports as a matter of right under FAVORS consent architecture.

6.4.6 The Maximization Imperative

The Erasure Completeness Principle replaces the illusion of perfect erasure with the discipline of honest maximization. It acknowledges that man cannot fully recycle all matter he creates, that information entropy is as real as thermodynamic entropy, and that the proper response to an infinite problem is not a false claim of finitude but a commitment to continuous, measured, transparent effort toward the best achievable state. The ECR does not promise completeness. It promises **maximum effort, honest measurement, and ongoing stewardship**—which is what trust actually requires.

7. GRACECHAIN: TRUST LEDGER AND MERITOCRATIC CREDENTIALING

Gracechain is the distributed trust ledger—not a financial ledger (that function is served by Meritcoin). Gracechain records, validates, and makes auditable the trust transactions that constitute the social fabric of NAC governance.

7.1 Trust Transaction Taxonomy

- **Credential Events.** Issuance, renewal, suspension, or revocation of SCALULAR certifications, IDIPITIS clearance levels, and EarnedPath milestones. Each event co-signed by credentialing authority and credential holder.
- **Consent Events.** FAVORS consent actions recorded using zero-knowledge proofs that confirm the action without revealing underlying biometric data.
- **Adjudicative Events.** Resolution Protocol outcomes, appeals, remands, compliance determinations. Linked to Public Reasoning Archive via CBGMODD provenance chains.
- **Institutional Transparency Events.** Mandatory disclosures: algorithmic audits, BERA index publications, resource allocation reports, governance decisions. Not voluntary—structural requirement.

7.2 Trust Score and the RCI

Gracechain maintains a rolling trust score for every entity (individual and institutional). The composite feeds directly into the BERA Resonant Continuity Index:

$$RCI = P_\Omega_norm \times ARI_sys \times VibConst$$

Where P_Ω_norm (normalized probability of systemic continuity) is substantially influenced by aggregate Gracechain trust scores. VibConst represents the qualitative systemic health of the governance environment—MIEVM Node 2 (Grok) correctly noted that this term requires rigorous, non-arbitrary determination methodology, which is acknowledged as a v0.95 deliverable.

8. ECONOMIC INTEGRATION: UBIMIA, MERITCOIN, AND THE US DOLLAR INTERFACE

The SPT architecture cannot function in an economic vacuum. Security infrastructure requires resources. Privacy enforcement requires ongoing operational expenditure. Trust maintenance requires sustained institutional investment. This section specifies the economic integration layer that connects SPT subsystems to the broader NAC financial architecture and to the existing United States dollar-denominated economy.

8.1 UBIMIA Baseline Provisioning

The Universal Basic Income Mechanism (UBIMIA) provides the economic floor that makes voluntary participation in the NAC system genuinely voluntary. Without material security, “informed consent” is a fiction—individuals under economic duress cannot freely choose whether to share biometric data, accept credential requirements, or participate in governance processes. UBIMIA ensures that no individual is economically coerced into SPT enrollment, data sharing, or credential pursuit.

UBIMIA baseline provisions are denominated in a dual-currency architecture: Meritcoin (the NAC internal unit of account) and the US dollar (the external interface currency).

The exchange mechanism between these two operates through Smart-Resonant Offset Contracts (SROC), which provide game-theoretic equilibrium between internal merit-based valuation and external market-based pricing.

8.2 US Dollar Interface and Constitutional Alignment

The SPT specification operates within, and is designed to complement rather than replace, the existing United States constitutional and financial architecture. Key alignment points:

- **Fourth Amendment Interface.** The FAVORS anti-surveillance provision operationalizes the Fourth Amendment's protection against unreasonable search and seizure in the biometric domain—extending constitutional protection to data categories that did not exist when the Amendment was ratified but fall squarely within its protective intent.
- **Fifth Amendment Interface.** IDIPITIS's transparent denial mechanism (explaining which clearance component fell short) operationalizes due process in the identity and access domain.
- **Fourteenth Amendment Interface.** NBERS's Immediate Rights Activation principle ensures equal protection under the NAC credential system from the moment of enrollment—no probationary second-class status.
- **Article I, Section 8 Interface.** Meritcoin and SROC operate within the congressional power to regulate commerce and coin money. The dual-currency architecture is designed to function under existing monetary law, not to circumvent it.

8.3 International Land Ownership Integration

The ILO (International Land Ownership) framework maps Personal/Public-Private stewardship classes to the foundational equation $C=R\times P/M$, creating a resource-governance layer that connects the SPT's digital architecture to the physical domain of land, property, and territorial jurisdiction. Within the United States, ILO stewardship classes are designed to align with existing federal, state, and municipal property law while extending the NAC governance principles to land-use decisions.

9. NBERS: NATIONAL BIOMETRIC ENROLLMENT AND RIGHTS SCAFFOLDING

NBERS is the enrollment and onboarding subsystem—the "front door" through which every person enters the NAC security-privacy-trust architecture. Its design carries extraordinary ethical weight.

9.1 Five Non-Negotiable Enrollment Principles

9. **Informed Voluntarism.** Enrollment must be genuinely voluntary. No government benefit, employment opportunity, or social service may be conditioned on NAC enrollment. UBIMIA provisions function in parallel with legacy systems during transition.
10. **Comprehensible Consent.** Conducted in the individual's primary language, at appropriate reading level, with mandatory opportunity for questions and withdrawal. Consent obtained through incomprehension is void.
11. **Minimum Viable Collection.** Only data strictly necessary for Tier 1 SSSC credentialing. Additional collection deferred to later lifecycle stages under progressive FAVORS consent layers.
12. **Immediate Rights Activation.** Full Tier 1 credential set and all UBIMIA baseline provisions upon enrollment completion. No probationary period, no "pending" status.
13. **Reversibility.** Withdrawal at any time via FAVORS Layer 7 across all data. CBGMODD proof-of-erasure triggered. Only anonymized withdrawal fact retained for demographic integrity.

10. DEGRADED STATE MANAGEMENT AND REMEDIATION

When any SPT Triad Integrity Condition component fails, the system enters a formally defined degraded state. Degraded states are cybernetic feedback signals, not emergencies by default.

10.1 Degradation Levels

Level	Designation	Trigger and Response
AMBER	**DRIFT**	SPT index within 10% of threshold. Automated alerts. Enhanced monitoring. No restrictions. 30-day remediation.
ORANGE	**BREACH**	Index(es) crossed threshold. Active remediation mandated. Enhanced audit mode. Resolution Protocol emergency provisions available. 14-day remediation.
RED	**COLLAPSE**	Multiple indices failed simultaneously or single index below 50% of baseline. Immediate GSSG tier escalation. Safe-mode. Planetary Reconciliation Council notified for Tier 4+. Immediate remediation.

10.2 Remediation Cycle

14. **Diagnose.** Identify degraded components, implicated threat classes, affected subsystems. MIEVM validation for complex cases.

15. **Contain.** Isolate affected domain. Containment evaluated against C=R×P/M (resource proportionality).
16. **Restore.** Return indices to baseline. Partial restoration acceptable with filed plan and timeline.
17. **Harden.** Structural changes to prevent recurrence. Evaluated against REAL=(E·M·R)/(T·S). Logged on Gracechain.

11. CROSS-SYSTEM DEPENDENCY GRAPH

The six SPT subsystems form a tightly coupled dependency graph in which failure of any one affects all others. MIEVM Node 2 (Grok) flagged this as a potential "brittle single-point-of-failure cascade." The assessment is acknowledged: the circular dependency is a deliberate architectural gamble that trades individual subsystem independence for comprehensive intrusion detection through systemic resonance. The risk is real; the mitigation strategy is the degraded state management system (Section 10), which is specifically designed to detect and contain cascading failures before they reach RED/COLLAPSE.

- **IDIPITIS** → **FAVORS** (biometric verification of identity claims)
- **FAVORS** → **CBGMODD** (provenance tracking and proof-of-erasure)
- **CBGMODD** → **Gracechain** (trust anchoring of provenance signatures)
- **Gracechain** → **SECUIR** (infrastructure integrity of trust ledger)
- **SECUIR** → **VERTECA** (energy resilience)
- **NBERS** → **ALL FIVE** (enrollment integration convergence point)

12. MIEVM META-VALIDATION

The SPT specification has been subjected to MIEVM analysis as documented in the Provenance Record. Ongoing ensemble validation is mandated at six-month intervals, with ad hoc validation triggered by any ORANGE or RED degradation event. Validation reports are appended to the Public Reasoning Archive and subject to CBGMODD provenance tracking.

MIEVM Node 2 (Grok) rated the specification 6.1/10 as a technical engineering blueprint; MIEVM Node 3 (DeepSeek) rated it 9.0/10 as an architectural vision and philosophical constitution. Both ratings are correct for their respective evaluation criteria. The gap between them—2.9 points—is the precise measure of the work that remains between architectural intent and implementation specification. The Patience Protocol (Section 13) governs the timeline for closing this gap.

13. PATIENCE PROTOCOL: VERSIONING DISCIPLINE

Per the MIEVM-validated dissertation on the necessity of patience in the architecture of trust, the following versioning discipline governs the SPT's path to normative status:

Version	Designation	Content	Target
v0.9	Architectural Intent	Philosophical foundation, subsystem roles, threat taxonomy, moral vector, MIEVM provenance. [CURRENT]	Mar 2026
v0.95	Consolidated Vision	+ Annotated bibliography mapping ERES to prior art (SSI, VCs, Hyperledger, IPFS, differential privacy) + glossary with precise variable definitions + VibConst determination methodology.	Q3 2026
v0.99	Reference Architecture	+ Protocol selections (cryptographic primitives, consensus mechanisms, DAG structures) + variable definitions with units/ranges + initial dependency modeling (TLA+ or Alloy).	Q1 2027
v1.0	Foundational Specification	+ Formal threat models with attack trees + governance constitution + proof-of-concept validation results + performance benchmarks.	2027–Q2 2028

The Patience Protocol affirms: premature declaration is its own class of threat. A specification that has not been tested against reality is not a specification—it is a hypothesis dressed in formal clothing. The ERES Institute will not rush. It will not pretend. It will build what can stand.

14. CLOSING DECLARATION

This specification recognizes a truth that most security architectures prefer to ignore: that security without privacy is oppression, that privacy without trust is isolation, and that trust without security is naïveté. The three must be held together, in dynamic tension, as a single system—a triad whose integrity is the precondition for every other governance function within the NAC framework.

The SPT architecture is offered as a formal draft—an MIEVM-consolidated architectural declaration of intent that maps the full scope of the protective challenge and proposes a unified response. It has been stress-tested by three independent analytical instruments and has incorporated their critiques honestly, including the uncomfortable ones. The gap between vision and implementation is measured, acknowledged, and scheduled.

The moral vector bears repeating: the purpose of security, privacy, and trust within the NAC framework is not to control people but to liberate them. To build a world in which individuals possess genuine sovereignty over their identities, their data, and their participation in governance. To ensure that institutions earn trust rather than demand it.

And to do all of this in service of the enduring imperative: ***Don't hurt yourself. Don't hurt others. Build for generations to come.***

APPENDIX A: A–Z NUMERICS INDEX

The following comprehensive index catalogues every formal term, formula, index, and subsystem referenced in this specification. Each entry carries an alphanumeric code (letter prefix = domain, two-digit suffix = sequence within domain) and cross-references to the section(s) in which the term is specified or substantially discussed.

Code	Term	Expansion	Description	§
A01	**ARI**	*Adaptive Resilience Index*	BERA index measuring system and individual resilience against adversarial action.	1.4, 3.2, 4, 9
A02	**Anti-Surveillance Provision**	*FAVORS absolute prohibition*	No passive biometric surveillance without explicit, informed consent. Non-waivable.	5.2
B01	**BERA**	*Biometric/Economic Resonance Architecture*	Four-index measurement system (ARI, ERI, RHC, RCI) governing all SPT assessments.	1, 4, 9
B02	**BIO_fresh**	*Biometric Freshness Function*	Time-decay variable in CLR_eff measuring recency of FAVORS verification.	3.2
C01	**C=R×P/M**	*Triune Math Formula 1*	Cybernetics = Resource × Purpose ÷ Method. Constraint validation for resource-proportionate action.	4.2, 10.2
C02	**CBGMODD**	*Stateful Data Taxonomy*	Data-chain integrity and provenance subsystem. Cryptographic DAG-based provenance chains.	6

C03	**CCAL v2.1**	*CARE Commons Attribution License*	Ethical open license governing all ERES publications.	Title, 14
C04	**CLR_eff**	*Effective Clearance Level*	Continuous clearance function: f(CRED_stack, ARI_current, BIO_fresh, RES_class).	3.2
C05	**CRED_stack**	*Credential Stack*	Cumulative credential weight from SCALULAR and Gracechain.	3.2
D01	**DAG**	*Directed Acyclic Graph*	Provenance chain structure used by CBGMODD for data-object biographies.	6.1
D02	**Degradation Levels**	*AMBER / ORANGE / RED*	Three-tier alert system for SPT Triad Integrity failures.	10
E01	**EarnedPath**	*EP=CPM×WBS+PERT*	Progressive milestone system for credential accumulation.	3.1
E02	**ECR**	*Erasure Completeness Ratio*	N_confirmed_destroyed / N_total_propagation_graph. Continuous metric (0–1) replacing binary erasure claims. Bounded by REAL constraint.	6.4
E03	**ECR_max**	*Maximum Achievable Erasure*	f(E, M, R, T, S). Thermodynamic ceiling on erasure completeness within finite resource envelope.	6.4.2
E04	**ERI**	*Economic Resilience Index*	BERA index measuring financial stability and UBIMIA distribution health.	1.4, 4, 8
E05	**ERES**	*Institute for New Age Cybernetics*	Founding institution (est. February 2012, Bella Vista, Arkansas).	Title
E06	**Erasure Completeness Principle**	*Thermodynamic Correction*	Erasure is asymptotic, not binary. Information entropy mirrors matter entropy. Governance response: maximization + honest measurement + ongoing stewardship.	6.4

F01	**FAVORS**	*Biometric Sovereignty Subsystem*	Seven-layer consent architecture governing all biometric data interactions.	5
F02	**FDRV**	*Foundational Document Reference Vector*	Cross-document linkage mechanism within the ERES corpus.	14
G01	**GAIA**	*Global Adaptive Intelligence Architecture*	Planetary-scale environmental and resource coordination system.	7.1
G02	**Gracechain**	*Trust Ledger*	Distributed trust ledger recording credential, consent, adjudicative, and transparency events.	7
G03	**GSSG**	*Global Smart Systems Governance*	Eight-tier planetary governance architecture (T1–T8).	1, 3.2, 10
H01	**Hardening**	*Post-Remediation Structural Change*	Fourth step of remediation cycle: REAL-validated systemic strengthening.	10.2
I01	**IDIPITIS**	*Identity Integrity & Security Clearance*	Primary boundary-enforcement subsystem. Identity lifecycle management.	3
I02	**ILO**	*International Land Ownership*	Personal/Public-Private stewardship framework mapped to C=R×P/M.	8.2
M01	**M×E+C=R**	*Triune Math Formula 2*	Matter × Energy + Constant = Reason. Constraint for material/energetic proportionality.	Anchor, 4.2, 10.2
M02	**Meritcoin**	*NAC Financial Instrument*	Digital currency operating alongside Gracechain trust ledger.	8
M03	**MIEVM**	*Multi-Instrument Ensemble Validation*	Parallel validation protocol using independent analytical instruments (AI nodes).	12
N01	**NAC**	*New Age Cybernetics*	Master framework encompassing all ERES subsystems. Est. 1997 (CyberRAVE era).	Title, 1

N02	**NBERS**	*National Biometric Enrollment & Rights Scaffolding*	Enrollment subsystem. Five principles: voluntarism, consent, minimum collection, immediate rights, reversibility.	9
P01	**PlayNAC**	*Governance Logic Substrate*	Procedural substrate for deliberation, mediation, and governance operations.	1.2
P02	**Proof-of-Erasure**	*CBGMODD Erasure Completeness Audit*	Replaced binary certificate with ECR-based completeness audit. Reports ratio, residual locations, containment measures. Thermodynamically honest.	6.2, 6.4
P03	**P_Ω_norm**	*Normalized Continuity Probability*	RCI component influenced by aggregate Gracechain trust scores.	1.3, 7.2
R01	**RCI**	*Resonant Continuity Index*	BERA index: RCI = P_Ω_norm × ARI_sys × VibConst. Systemic trust measurement.	1.3, 1.4, 7.2
R02	**REAL=(E·M·R)/(T·S)**	*Triune Math Formula 3*	Energy·Matter·Resonance over Time·Space. Spatiotemporal implementability constraint.	4.2, 10.2
R03	**RHC**	*Resonant Harmony Cycle*	BERA index measuring community-level informational self-determination.	1.2, 1.4
R04	**ResProtocol**	*Resolution Protocol*	Formal dispute resolution specification (ERES-RESPROTOCOL-2026-001).	10, 14
S01	**SCALULAR**	*Certification Architecture*	Scalable Certification Architecture for Lifelong Universal Learning and Adaptive Resilience.	3.1, 9
S02	**SECUIR**	*Infrastructure Protection*	Five-ring defense-in-depth subsystem (Perimeter, Transport, Compute, Storage, Energy).	7

S03	**SOMT**	*Sociocratic Overlay Metadata Tapestry*	Coordination and transparency network for all NAC governance communications.	6.3, 7.1
S04	**SPT Triad Integrity Condition**	*Formal Health Metric*	Three simultaneous BERA thresholds defining system health (ARI ≥ floor, RHC ∈ [μ±1σ], RCI ≥ 0.7×baseline).	1.4
S05	**SROC**	*Smart-Resonant Offset Contracts*	Game-theoretic economic instruments within NAC financial architecture.	8.2
S06	**SSSC**	*Solid-State Smart-City Citizen*	Tier 1 baseline credential issued upon NBERS enrollment.	3.1, 9
T01	**Threat Classes I–V**	*Unified Threat Model*	Identity Compromise, Data Exfiltration, Trust Erosion, Infrastructure Disruption, Epistemic Manipulation.	2
T02	**Triune Math**	*Three Canonical Formulas*	C=R×P/M; M×E+C=R; REAL=(E·M·R)/(T·S). Axiomatic constraint set.	4.2
U01	**UBIMIA**	*Universal Basic Income Mechanism*	Economic provisioning system ensuring baseline material security for all NAC citizens.	8, 9
V01	**VERTECA**	*Energy Architecture*	Vertical Energy Resonance Temporal Environmental Cybernetic Architecture.	7.2
V02	**VibConst**	*Vibrational Constant*	RCI component representing qualitative systemic health. Requires rigorous non-arbitrary determination.	1.3, 7.2, 12
Z01	**Zero-Knowledge Proofs**	*Privacy Primitive*	Cryptographic technique used by Gracechain for consent event logging without revealing biometric data.	7.1

APPENDIX B: REFERENCES

B.1 ERES Internal Corpus

18. Sprute, J. A. (2026). ERES Security, Privacy and Trust Specification (v0.9 DRAFT). ERES-SPT-2026-001.
19. Sprute, J. A. (2026). ERES Resolution Protocol (v1.0). ERES-RESPROTOCOL-2026-001.
20. Sprute, J. A. (2026). Book One: ONE GOOD — Synthetic AI Constitution Through UBIMIA.
21. Sprute, J. A. (2026). Book Two: SECURITY-CLEARANCE — IDIPITIS Framework for NBERS.
22. Sprute, J. A. (2026). Book Three: DATA-INTEGRITY — FAVORS for CBGMODD GAIA SOMT.
23. Sprute, J. A. (2026). ERES Complete Description-Corpus (491 paragraphs, 12 parts).
24. Sprute, J. A. (2026). ESVRD v4.0. Empirically grounded propulsion white paper.
25. Sprute, J. A. (2026). FAVORS-BERA v6.0. Submitted to DBDM 2026 (Sydney).

B.2 External and Foundational Literature

26. Wiener, N. (1948). Cybernetics: Or Control and Communication in the Animal and the Machine. MIT Press.
27. Ashby, W. R. (1956). An Introduction to Cybernetics. Chapman & Hall.
28. Beer, S. (1972). Brain of the Firm: The Managerial Cybernetics of Organization. Allen Lane.
29. Schneier, B. (2015). Data and Goliath. W. W. Norton.
30. Zuboff, S. (2019). The Age of Surveillance Capitalism. PublicAffairs.
31. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System.
32. Allen, C. et al. (2016). The Sovrin Protocol for Self-Sovereign Identity. Sovrin Foundation.
33. Sporny, M. et al. (2019). Verifiable Credentials Data Model 1.0. W3C Recommendation.
34. Benet, J. (2014). IPFS — Content Addressed, Versioned, P2P File System.
35. Butzbach, J. D. (2025). Continuity Theory. [Co-developed RCI specification with Sprute.]

END OF DOCUMENT

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